$\Rightarrow$ Нет сильной регулярности

№ 6

$$\xi \sim p(x) = \begin{cases} \frac{\theta-1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases} \qquad \theta > 1$$

а) $L(\theta) = \frac{(\theta-1)^n}{(\prod x_i)^{\theta}}$

$\ln L(\theta) = \ln(\theta-1)^n - \theta \ln(\prod x_i)$

Если мн-во открыто, то можно пользоваться безусловной оптимизацией

$(\ln L(\theta))'_{\theta} = n \frac{1}{\theta-1} - \ln(\prod x_i) = 0$

$$n = (\theta - 1)\ln\left(\prod x_i\right)$$

$$\tilde{\theta} = 1 + \frac{n}{\ln\left(\prod x_i\right)}$$

смещ. оценка

$$\left(\ln L(\theta)\right)''_{\theta\theta} = \frac{-n}{(\theta-1)^2} < 0$$

при $\theta > 1$

$\Longrightarrow$ максимум

$$\tilde{\theta} = 1 + \frac{n}{\sum_{i=1}^{n} \ln x_i}$$

5. Доверительный интервал для медианы:

Точкой:

медиана $\int_1^{x_m} \frac{\theta - 1}{x^\theta}\, dx = \frac{1}{2}$

$$-\frac{\cancel{(\theta-1)}}{\cancel{(\theta-1)}} \frac{1}{x^{\theta-1}} \Big|_1^{x_m} = \frac{1}{2}$$

$$-\frac{1}{x_m^{\theta-1}} + 1 = \frac{1}{2} \Longrightarrow x_m^{\theta-1} = 2$$

$$\cancel{X_m = 2^{\frac{1}{1-\theta}}} \quad X_m = 2^{\frac{1}{\theta-1}}$$

$\xi \sim F(x)$ - нагадём:

$$F_\xi(x) = \int_{-\infty}^{x} P(x)\,dx = \int_{1}^{x} \frac{\theta-1}{x^\theta}\,dx =$$

$$= (\theta-1)\int_{1}^{x} x^{-\theta}\,dx = \frac{(\theta-1)}{-\theta+1} x^{-\theta+1} \Big|_{1}^{x} =$$

$$= -1\left(x^{-\theta+1} - 1\right) = 1 - x^{-\theta+1}$$

Рассмотр. $\eta = \ln \xi$

$$\cancel{F_\eta = P(}$$

$$F_\eta(y) = P(\eta < y) = P(\ln \xi < y) =$$

$$= P(\xi < e^y) = 1 - (e^y)^{-\theta+1} =$$

$$= 1 - e^{(1-\theta)y}$$

$$P_\eta(y) = (\theta-1)\, e^{-(\theta-1)y} = \Gamma((\theta-1), 1)$$

$\Gamma(1) = 1$

$$\zeta = \sum \eta_i = \sum \ln \xi_i \sim \Gamma((\theta-1), n)$$

по св-ву суммы

Избавимся от θ:

$$\gamma = 2(\theta-1)\zeta = 2(\theta-1)\underbrace{\sum \ln \xi_i}_{T} = 2(\theta-1)T$$

$$F_\gamma(z) = P(\gamma < z) = P(2(\theta-1)\zeta < z) =$$

$$= P\left(y < \frac{z}{2(\theta-1)}\right) \cancel{= P_\zeta}$$

$$P_\gamma(z) = P_\gamma\left(\frac{z}{2(\theta-1)}\right)\underbrace{\left|\frac{\partial y}{\partial z}\right|}_{\text{Якобиан в интеграле}} =$$

$$= \frac{\cancel{(\theta-1)^n}}{\Gamma(n)} \frac{z^{n-1}}{2^{n-1}\cancel{(\theta-1)^{n-1}}} e^{-\frac{\cancel{(\theta-1)}z}{2\cancel{(\theta-1)}}} \frac{1}{2\cancel{(\theta-1)}} =$$

$$= \underbrace{\frac{z^{n-1}}{\Gamma(n)2^n} e^{-\frac{z}{2}}}_{\frac{(\frac{1}{2})^n}{\Gamma(n)} z^{(n-1)} e^{-\frac{z}{2}}} \sim \Gamma\left(\frac{1}{2}, n\right)$$

серьёзно ёкает, да это Гамма!

$$t_1 = \Gamma\left(\frac{1}{2}, n\right)_{\cancel{0,025}\ \frac{1-\beta}{2}}$$

$$t_2 = \Gamma\left(\frac{1}{2}, n\right)_{\cancel{0,975}\ \frac{1+\beta}{2}}$$

$$t_1 \leq 2(\theta - 1)T \leq t_2$$

$$\frac{t_1}{2T} \leq \theta - 1 \leq \frac{t_2}{2T}$$

$$\left(\frac{t_2}{2T}\right)^{-1} \leq \frac{1}{\theta - 1} \leq \left(\frac{t_1}{2T}\right)^{-1}$$

$$2^{\left(\frac{t_2}{2T}\right)^{-1}} \leq X_M \leq 2^{\left(\frac{t_1}{2T}\right)^{-1}}$$

Асимптотический:

Проверим состоятельность мл-вх и сильную регулярность:

$\theta > 1$ – скрытое

Для сильной регулярности:

$$\int_{A(\theta)} \frac{\partial^2}{\partial \theta^2} P(x, \theta)\, dx + \sum_k \frac{\partial^2}{\partial \theta^2} P_k(\theta) = 0$$

$$p(x) = (\theta-1)x^{-\theta} = \theta x^{-\theta} - x^{-\theta}$$

$$\frac{\partial p}{\partial \theta} = x^{-\theta} + \theta x^{-\theta} \ln x \,(-1) + x^{-\theta} \ln x$$

$$\frac{\partial^2 p}{\partial \theta^2} = -x^{-\theta} \ln x - x^{-\theta} \ln x + \theta x^{-\theta} \ln^2 x - x^{-\theta} \ln^2 x$$

$$\int_1^{+\infty} \frac{\partial^2 p}{\partial \theta^2} = \{ u = \ln x \} = \int_0^{+\infty} (-2e^{-(\theta-1)u} \cdot u + (\theta-1) e^{-(\theta-1)u} \cdot u^2) du =$$

$$= \left\{ \int_0^{\infty} u e^{-\lambda u} du = \frac{1}{\lambda^2} \int_0^{\infty} u^2 e^{-\lambda u} du = \frac{2}{\lambda^3} \right\} =$$

$$= -2 \cdot \frac{1}{\lambda^2} + \lambda \cdot \frac{2}{\lambda^3} = 0$$

$\Rightarrow$ модель только разумерна.

$$\sqrt{n} \frac{g(\tilde{\theta}) - g(\theta)}{\sigma(\theta)} \rightsquigarrow N(0,1)$$

$$g(\theta) = 2^{\frac{1}{\theta-1}} \qquad \nabla g(\theta) = \frac{2^{\frac{1}{\theta-1}} \ln 2}{-(\theta-1)^2}$$

Инф. Фишера:

$$I(\theta) = \int_1^{+\infty} \left(\frac{\partial \ln p}{\partial \theta}\right)^2 p \, dx =$$

$$= \left\{ \begin{array}{l} \ln p = \ln \frac{\theta - 1}{x^\theta} = \ln(\theta - 1) - \theta \ln x \\ \frac{\partial \ln p}{\partial \theta} = \frac{1}{(\theta - 1)} - \ln x \end{array} \right\} =$$

$$= \int_1^{+\infty} \left(\frac{1}{(\theta-1)} - \ln x\right)^2 \frac{\theta - 1}{x^\theta} \, dx =$$

$$= \int_1^{+\infty} \left(\frac{1}{(\theta-1)^2} - \frac{2 \ln x}{(\theta-1)} + \ln^2 x\right) \left(\frac{\theta - 1}{x^\theta}\right) dx =$$

$$= \int_1^{+\infty} \left(\frac{1}{(\theta-1)\, x^\theta} - \frac{2 \ln x}{x^\theta} + \frac{\ln^2 x}{x^\theta}\right) dx =$$

$$= \left\{ u = \ln x \quad \lambda = (\theta - 1) \right\} =$$

$$= \lambda \left(\frac{1}{\lambda^2} \int_0^\infty e^{-\lambda u} \, du - \frac{2}{\lambda} \int_0^\infty u e^{-\lambda u} \, du + \int_0^\infty u^2 e^{-\lambda u} \, du \right) =$$

$$= \left\{ \begin{array}{l} \int_0^\infty u^k e^{-\lambda u} \, du = \frac{k!}{\lambda^{k+1}} \\ \int_0^\infty e^{-\lambda u} \, du = \frac{1}{\lambda} \quad \int_0^\infty u e^{-\lambda u} \, du = \frac{1}{\lambda^2} \quad \int_0^\infty u^2 e^{-\lambda u} \, du = \frac{2}{\lambda^3} \end{array} \right\}$$

$$I(\theta) = \frac{1}{\lambda^2}$$

$$I = \frac{1}{(\theta-1)^2} > 0 \text{ и непр}$$

(модель регулярна)

$$\sigma^2(\theta) = \nabla^T g(\theta)\, I^{-1}(\theta)\, \nabla g(\theta) =$$

$$= (\theta-1)^2 \left(\frac{2^{\frac{1}{\theta-1}} \ln 2}{(\theta-1)^2} \right)^2 = \frac{2^{\frac{2}{\theta-1}} \ln^2 2}{(\theta-1)^2}$$

$$\sqrt{n}\, \frac{\cancel{X_m}\, g(\tilde\theta) - X_m}{2^{\frac{1}{\tilde\theta-1}} \ln 2 / (\tilde\theta-1)} \rightsquigarrow N(0,1)$$

пусть $\tilde X_m = 2^{\frac{1}{\tilde\theta-1}}$

$$u_{\frac{1-\beta}{2}} \le \sqrt{n}\, \frac{(\tilde X_m - X_m)\cdot(\tilde\theta-1)}{\tilde X_m \ln 2} \le u_{\frac{1+\beta}{2}}$$

$$\frac{u_{\frac{1-\beta}{2}} \cdot \ln 2 \cdot \tilde X_m}{(\tilde\theta-1)\sqrt{n}} \le \tilde X_m - X_m \le \frac{u_{\frac{1+\beta}{2}} \cdot \ln 2\, \tilde X_m}{(\tilde\theta-1)\sqrt{n}}$$

$$\tilde X_m - \frac{u_{\frac{1+\beta}{2}} \ln 2\, \tilde X_m}{(\tilde\theta-1)\sqrt{n}} \le X_m \le \tilde X_m - \frac{u_{\frac{1-\beta}{2}} \ln 2\, \tilde X_m}{(\tilde\theta-1)\sqrt{n}}$$

c.) $g(\theta) = \theta \quad \nabla g(\theta) = 1$

$$\sigma^2(\theta) = 1 \cdot (\theta - 1)^2 \cdot 1 = (\theta - 1)^2$$

$$\sqrt{n}\,\frac{\tilde{\theta} - \theta}{(\tilde{\theta} - 1)} \rightsquigarrow N(0,1)$$

$$u_{\frac{1-\beta}{2}} \leq \sqrt{n}\,\frac{\tilde{\theta} - \theta}{\tilde{\theta} - 1} \leq u_{\frac{1+\beta}{2}}$$

$$\frac{1}{\sqrt{n}} u_{\frac{1-\beta}{2}} (\tilde{\theta} - 1) \leq \tilde{\theta} - \theta \leq \frac{1}{\sqrt{n}} u_{\frac{1+\beta}{2}} (\tilde{\theta} - 1)$$

$$\tilde{\theta} - \frac{1}{\sqrt{n}} u_{\frac{1+\beta}{2}} (\tilde{\theta} - 1) \leq \theta \leq \tilde{\theta} - \frac{1}{\sqrt{n}} u_{\frac{1-\beta}{2}} (\tilde{\theta} - 1)$$